

Research

Comprehensive genome-wide identification and analysis of the glycolate oxidase (*GLO*) gene family in tomato (*Solanum lycopersicum* L.) during biotic stress and fruit development stages

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Abstract

GLO/GOX represents an essential enzyme in photorespiratory metabolism in plants, catalyzing glycolate oxidation to glyoxylate and producing H_2O_2 , a signaling molecule involved in various physiological processes and defense mechanisms. Here, we identified and analyzed six putative glycolate oxidase genes in tomatoes and classified them into three groups based on phylogenetic analysis, gene structure, and motif composition. Our data showed that tandem duplication contributed mainly to expanding the GLO gene families in tomatoes. The promoter analysis identified the specific cis-regulatory elements (CREs) related to stress, light, hormone, and development. Gene ontology analysis found that all six *SIGLO* genes are mainly associated with oxidoreductase activity. Transcriptome analysis of the interactions of tomatoes with AMF emphasized the role of *SIGLO* genes in fruit nutrition improvement. The gene expression varied among different *SIGLO*s. Specifically, *SIGLO2* showed the lowest expression, while *SIGLO1* and *SIGLO3* were highly expressed in red fruits; *SIGLO6* was highly expressed in green but unripe in red fruits. Alternatively, *SIGLO4* and *SIGLO5* were the highest, indicating involvement in the potential amelioration of fruit quality. Gene expression analysis using qRT-PCR showed that *SIGLO*s are differentially expressed across different organs and at various stages of fruit development, implicating this gene family in plant development and physiology. Furthermore, RNA-seq comparison under biotic stress demonstrated differential expression within different tomato cultivars. *SIGLO1* and *SIGLO2* were moderately expressed, while *SIGLO3* expression decreased in infected roots. *SIGLO4* and *SIGLO6* were upregulated within the resistant Mogeor cultivars, indicating their involvement in pathogen-associated transcriptional regulation. Overall, our study demonstrates the multifaceted role of *SIGLO* genes in stress response, development, and fruit quality improvement.

Keywords Tomato · Glycolate oxidase · *SIGLO* genes · Phylogenetic analysis · Plant stress

Muhammad Zeshan Haider and Adnan Sami have equal contribution.

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1 Introduction

Glycolate oxidase (GOX/GLO) is an ancient peroxisomal eukaryotic enzyme involved in photorespiratory metabolism [42]. It is a member of the superfamily of (L)-2-hydroxy acid-oxidases ((L)-2-HAOX) [14, 25, 29]. It participates in the oxidation of various 2-hydroxy acid substrates, such as catalyzing the transformation of glycolate into glyoxylate while generating hydrogen peroxide (H_2O_2) [90].

GOX/GLO short-chain α -hydroxy-acid oxidases are flavin mononucleotide (FMN)-dependent enzymes, where FMN acts as a cofactor in oxidation-reaction modulating reactive oxygen species (ROS) mediated signal transduction in plants [51]. The reaction is divided into two significant steps: in the first half, glycolate is oxidized to form glyoxylate by FMN, which is a two-electron transfer process, whereas, in the second half, FMN is re-oxidized by O_2 , yielding H_2O_2 [52, 53]. Therefore, GLO is considered an essential engine of H_2O_2 production, accounting for approximately 70% of total H_2O_2 in C3 plants, and also serves as a significant signaling molecule [50].

GLOs have been identified and characterized in several plant species, such as *Arabidopsis* [42, 67] *tobacco* [90], *spinach* [16], *maize* [62], and *rice* [50]; they are reported to play various roles in plant physiological processes, photosynthetic regulation, and biotic and abiotic stress resistance [90]. GLO strongly regulates photosynthesis via feed-back inhibition on Rubisco activase; suppressing GLO results in glyoxylate accumulation, halting photosynthesis and resulting in stunted rice growth, whereas over-expressing GLO exhibits improved photosynthesis activity [43, 89]. In addition, the induced activity of GLO has been frequently observed in response to environmental stresses such as high temperature and light [22, 68], drought, cold assimilation, and freezing [5, 7, 19]. Furthermore, H_2O_2 and salicylic acid (SA) activity have been correspondingly induced upon elevated levels of GLO, suggesting the critical role of GLO in triggering stress defense response [22, 31]. In spinach, *SoGLO2* and *SoGLO3* are involved in oxalate biosynthesis under normal conditions [16]. Interestingly, GLO-derived H_2O_2 demonstrated a paramount role in gene-to-gene and non-host resistance in *Arabidopsis thaliana* and *Nicotiana benthamiana* [67]. Intriguingly, GOX proteins were abundantly observed in resistant *B. napus* plants exhibiting tolerance against *Leptosphaeria maculans* [10, 15]. Similar studies were performed in barley and melon against *Bipolaris sorokiniana* and *Pseudoperonospora cubensis* [74, 83]. Moreover, GLO is found to be involved in *barley stripe mosaic virus* infection in barley and basal defense against *Pseudomonas syringae* in tomatoes [1, 59, 91]. Alban et al [46] showed that GOX2 in DspA/E triggers cell death and non-host resistance against *E. amylovora* [46, 69].

Tomato (*Solanum lycopersicum*) is a commercially valuable horticulture crop and a model plant for genetic and genomic research, including ripening mechanisms and stress studies [82, 85]. According to the Food and Agriculture Organization (FAO) of the United Nations (UN) statistics, it is the third most widely consumed vegetable, with a global production of 187 million metric tonnes (MMT) in 2020 (<http://www.fao.org/faostat/en/#data/QC>, [2, 81]). Consumers favor it due to its abundant nutritional value and unsurpassable carotenoids, especially beta-carotene and antioxidants, vitamins A, B, C, and E; nicotinic acid and lycopene [2, 71, 80, 82]. However, its production and quality are highly challenged by various biotic stresses. Several studies have demonstrated the function and characterization of GLO in different plants. However, there is a need to uncover the genome-wide mechanisms of GLO proteins in tomatoes against biotic resistance.

In the present study, we utilized bioinformatics methods to identify and comprehensively characterize *SIGLOs* and analyze their physiochemical properties, evolutionary relationship with other plant species, synteny relationship, conserved domain architecture, chromosomal locations, and *cis*-regulatory elements in the promoter region of the *GLO* gene family of *Solanum lycopersicum*. The expression characteristics of *SIGLOs* in different tissues under biotic stress will also provide a theoretical foundation for exploring their potential functions.

2 Materials and methods

2.1 Sequence retrieval and characterization

The Pfam database retrieved the conserved sequence for the FMN-dependent dehydrogenase (FMN_dh) domain sequence (PF01070) [57]. The domain sequence was used to identify *GLO* genes in the tomato genome from the Phytozome v13 database by employing the BLAST-P algorithm with the E value $\leq 1 \times 10^{-5}$ [34]. The retrieved protein sequences were then subjected to the National Center for Biotechnology Information (NCBI) Conserved Domain Database (CDD) to confirm domain presence. Protein sequences lacking the FMN_dh domain were excluded [54]. The molecular weight, the number of amino acids, and the isoelectric point (pI) were determined using ProtParam and Expasy-Compute pI/Mw

[32, 73]. The nuclear localization was predicted using the online server NLSdb [12]. The chromosomal location, accession numbers, and sequence information were retrieved from the Phytozome database [34, 70].

2.2 Multiple sequence alignment, phylogenetic analysis, and motif analysis

The amino acid sequences of six GLO proteins from *S. lycopersicum* (tomato), five GLO proteins from *A. thaliana* (Arabidopsis), five GLO from *Spinacia oleracea* (spinach), and four GLO from *Daucus carota* (carrot) were aligned using MUSCLE in MEGA X [27]. With the MEGA 11.0 version, the phylogenetic tree was constructed using the neighbor-joining (NJ) method, and the tree's dependability was assessed using default settings of pairwise deletion and 1000 iterations of bootstraps [84]. *SIGLO*, *AtGLO*, *SoGLO*, and *DcGLO* protein sequences were subjected to Multiple EM for motif elicitation (MEME Suite version 5.4.1) to identify the conserved motifs [11, 77].

2.3 Gene structure analysis and chromosomal mapping

Using the Gene Structure Display Server (GSDS 2.0), a schematic diagram of the gene structure of *SIGLO* genes was constructed [38]. The chromosomal locations of *SIGLO*s were retrieved from the Phytozome database and visualized using TB tools [20, 58].

2.4 Gene duplication and synteny analysis

Gene duplication events were analyzed using the Multiple Collinearity Scan toolkit (MCScanX) with default settings [21, 40]. To determine the time of divergence of the glycolate oxidase gene family in tomatoes, the synonymous (Ks) and nonsynonymous (Ka) substitution rates were estimated by using the Simple Ka/Ks Calculator (NG) in TBtools [20]. The Ks values were subsequently utilized to approximate date the duplication events according to $T = Ks/2\lambda$, where the value of λ for tomato = 1.5×10^{-8} substitutions/synonymous site/year [4, 17, 36].

The tandem and segmental duplication events were analyzed using the Multiple Collinearity Scan toolkit (MCScanX) with default parameters for synteny analysis. The synteny relationships between the paralogous *GLO* genes in tomatoes were visualized using the Advanced Circos module in TBtools [20, 39]. Three synteny analysis maps were constructed to examine the orthologous gene conservation of *SIGLO*s with three other plant species for dual synteny analysis. This was accomplished using the Dual Synteny Plotter in TBtools [6, 20].

2.5 Prediction of Cis-regulatory elements

To predict plant *cis*-regulatory elements in the promoter regions, a 1000-bp sequence upstream from the initiation codon of each putative *SIGLO* gene was retrieved from the tomato genome (*Solanum lycopersicum* ITAG4.0) via the Phytozome database. The *cis*-elements within these promoter sequences were then identified using the PlantCARE database (<https://bioinformatics.psb.ugent.be/webtools/plantcare/html/>) [48].

2.6 Subcellular localization prediction and gene ontology analysis

Subcellular localization of *SIGLO* genes was predicted with the help of an online tool, WoLF PSORT, available at (<https://wolfpsort.hgc.jp/>) [8, 37]. Gene Ontology (GO) term enrichment analysis was performed to investigate the participation of *SIGLO* genes in various molecular functions and biological processes. The ontology information was retrieved from the Phytozome database [13, 34, 65]. Moreover, a dot plot displaying other vital biological processes associated with the *SIGLO* genes was constructed to visualize the enriched pathways and the number of genes involved. This was accomplished by implementing ShinyGO 0.76 (<http://bioinformatics.sdstate.edu/go/>) [3, 33].

2.7 Transcriptome analysis

The expression profile of the tomato glycolate oxidase genes was investigated using the RNA-seq data obtained from the Expression Atlas database (<https://www.ebi.ac.uk/gxa/>) [60]. The expression was analyzed in vegetative tissue (apical meristem), reproductive tissues (flower bud, seed, and pericarp), and fruits at two stages from mycorrhizal and non-mycorrhizal plants. The expression profile of *SIGLO* genes in root tissues under biotic stress conditions was also analyzed.

The RNA-seq data for each particular tissue was obtained from experiments available at Expression Atlas [60] for the gene expression analysis. The RPKM values were log2 transformed, and expression patterns were visualized as Heat maps generated in the HeatMap Illustration in TBtools [20].

2.8 MicroRNA targets prediction

The putative microRNA target sites analysis was performed to predict the miRNAs that could most likely target the tomato *GLO* genes. Six *SIGLO* genes were searched for sequences complementary to miRNAs at psRNATarget (<https://www.zhaolab.org/psRNATarget/>) with default parameters [23].

2.9 RNA extraction and cDNA synthesis

The total RNA was extracted from the Tomato (*Solanum lycopersicum*) cultivar 'Micro-Tom' using TRIzol[™] reagent. Grind the tissues mechanically using Liquid nitrogen. All the samples were taken in the Eppendorf tube containing 1 ml of TRIzol[™] reagent followed by the addition of 200 µl chloroform. The samples were homogenized and incubated on ice for 3–5 min. After incubation, they were centrifuged at 12,000 rpm for 15 min at 4 °C. The supernatant was then transferred to a new Eppendorf tube, and an equal amount of chilled isopropanol was added. The samples were incubated on ice for 10 min and centrifuged at 12,000 rpm for 10 min. The supernatant was discarded, and the pellet was obtained. The pellet was washed with 70% ethanol and centrifuged at 7500 rpm for 10 min at 4 °C. The supernatant was discarded, and the pellet was air-dried for 15–30 min. Then, 50–60 µl RNase-free water was added, and the pellet was stored at –80 °C until use. According to the manufacturer's protocol, the cDNA was synthesized using TOROBlue[®] qRT PreMix with gDNA Eraser 2.0. 2 ng/µL RNA was treated with 5 × gDNA Eraser at 37 °C for 5 min, followed by 65 °C for 2 min. Then, 5xqRT premix II and reverse transcript enzyme mix were used. RT reaction was conducted using 37 °C for 15 min and 98 °C for 2 min and was kept 4 °C.

2.10 Semi-quantitative PCR

Semi-quantitative PCR was performed using the following protocol. cDNA was used as a template. The PCR reaction mixture was prepared accordingly: 2X M5 Taq HiFi PCR mix (with blue dye) 10 µL, forward qPCR primer (20 ng/µL), forward qPCR primer (20 ng/µL), dH₂O was to make a total volume of 20 µL. The PCR reaction was conducted according to the following profile: Initial denaturation of 95 °C for 10 min, 30 cycles of denaturation of 95 °C for 30 s, annealing 55 °C for 30 s, extension of 72 °C for 20 s; and final extension at 72 °C for 10 min. Actin7 was used as a housekeeping gene.

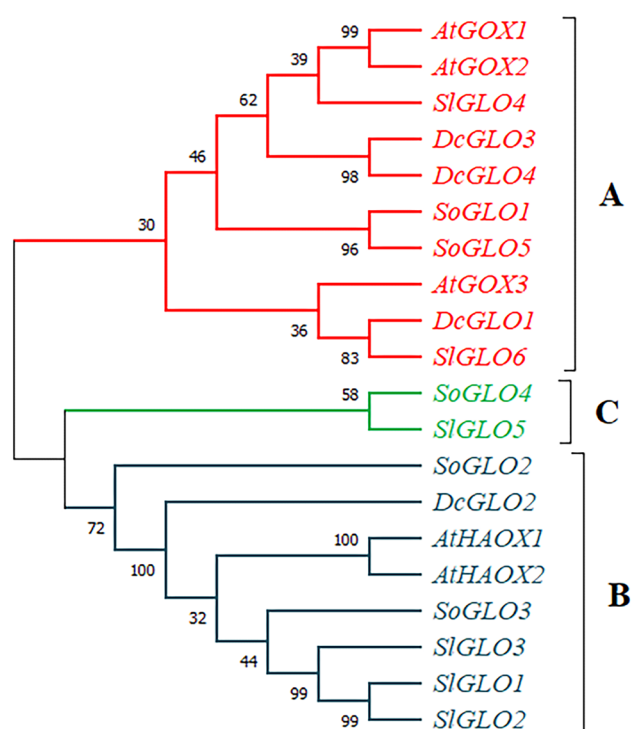
2.11 Quantitative PCR

qPCR was performed using TOROGreen[®] qPCR Master Mix. The PCR was conducted using a 2-step profile per the manufacturer's protocol. The qPCR was performed in three replicates under conditions of pre-denaturation at 95 °C for 60 s, followed by 40 cycles of denaturing at 95 °C for 10 s and annealing/extension at 60 °C for 30 s. Quantification was performed using the 2^{−ddCT} method and was normalized through the quantity of actin7 as a reference gene. The sequences of all the gene-specific primers are enlisted in supplementary data. The forward and reverse primer of the corresponding gene ID are written below.

Gene ID	Forward primer	Reverse primer
SIGLO1	GAGGAATTAGGCGAGGAACAG	TTGCTTTACCCAGATTCTCC
SIGLO2	AGCTGTTCTGATTGGTCGAC	TGGCTTCTGGTAATGTCATCC
SIGLO3	GAGGCGAGGAACAGACATATTC	CCCTGAGCATTTGAATAACCTG
SIGLO4	TCCAGACTATCACTTCAATGCC	TGTTGCAGGGACATAATCGAG
SIGLO5	TCGTGAAAAGAGCAGAAAGGG	TCCAAGATCCAGTCTTCAAAG
SIGLO6	GCTCTTACAGTTGATACCCCG	ATATGAAGCCAATCCAGAGTCC
ACT7	GGTATCCACGAGACTACCTACA	TGCTCATACGGTCAGCAATAC

Table 1 List of *GLO* genes identified in tomato (*Solanum lycopersicum*) along with their details about chromosomal location, peptide size, pI (isoelectric point), Mw (molecular weight), and NLS signals

Gene name	Gene ID (<i>Solanum lycopersicum</i> ITAG4.0)	Chr. no	Chr. location	Direction	Peptide (AA)	pI	Mw/1000 (kDa)	NLS signal
<i>SIGLO1</i>	Solyc03 g122130.3.1	03	64516138..64519356	Forward	364	7.56	39.90444	N/A
<i>SIGLO2</i>	Solyc03 g122140.3.1	03	64519796..64522737	Reverse	363	7.64	39.4076	N/A
<i>SIGLO3</i>	Solyc03 g122170.4.1	03	64527920..64532891	Forward	364	6.76	39.85733	N/A
<i>SIGLO4</i>	Solyc07 g056540.3.1	07	64247198..64252056	Forward	371	8.98	40.60697	N/A
<i>SIGLO5</i>	Solyc08 g048250.4.1	08	14000208..14007585	Forward	230	9.04	24.96202	N/A
<i>SIGLO6</i>	Solyc10 g007600.3.1	10	1777418..1781989	Reverse	371	9.01	40.68397	N/A

Fig. 1 Phylogenetic analysis of *GLO* genes in *S. lycopersicum*, *A. thaliana*, *S. oleracea*, and *D. carota*. The phylogenetic tree was constructed by aligning the full-length sequences of six tomato, five Arabidopsis, five spinach, and four carrot glycolate oxidase proteins using MEGA11 software by the Neighbor-Joining method and with 1000 bootstrap replicates [84]. The genes in the three groups, A, B, and C, are indicated by red, blue, and green colors, respectively

3 Results

3.1 Identification and characterization of *GLO* gene family in tomato

A BLASTp was used to identify the *GLO* gene family in tomatoes using the FMN-dependent dehydrogenase domain (PF01070), and six *GLOs* were identified against the tomato genome '*S. lycopersicum* ITAG4.0' available at Phytozome v13. These six *SIGLO* genes were renamed according to the order of their physical positions on chromosomes (Table 1). The physio-chemical analysis predicted that the molecular weight of the *GLO* gene family members in tomatoes ranged from 24.96 kDa to 40.68 kDa, while their lengths varied from 230 to 371 amino acids. The six proteins' isoelectric points (pI) ranged from 6.76 to 9.04. Moreover, none of the six *SIGLO* genes showed Nuclear Localization Signals (NLS) as predicted by NLSdb (Table 1).

3.2 Comparative phylogenetic analysis of *GLO* gene family

The phylogenetic analysis showed that these twenty *GLO* proteins could be divided into three distinct groups: A, B, and C (Fig. 1). Group A is comprised of a total of ten *GLO* proteins, of which two were tomato (*SIGLO4* and *SIGLO6*), three of Arabidopsis *GOX* proteins (*AtGOX1*, *AtGOX2*, and *AtGOX3*), whereas, the remaining are of spinach *GLOs* (*SoGLO1* and *SoGLO5*) and carrot *GLOs* (*DcGLO1*, *DcGLO3*, and *DcGLO4*), respectively. 'Group B' consists of eight genes in total, including three tomato genes (*SIGLO1*, *SIGLO2*, and *SIGLO3*), two Arabidopsis genes (*AtHAOX1* and *AtHAOX2*), two spinach genes (*SoGLO2* and *SoGLO3*) and one carrot gene (*DcGLO2*). 'Group C' contains only two *GLO* genes from tomato (*SIGLO5*) and the other from spinach (*SoGLO4*). Since proteins in common clad exhibit similar structures and functions, it can be inferred that all the *GLO* genes in similar clades might have structural and functional similarities (Fig. 1).

3.3 Gene structure and motif analysis

Five *SIGLO* genes, *SIGLO1*, *SIGLO2*, *SIGLO3*, *SIGLO4*, and *SIGLO6*, have ten introns, whereas one gene, *SIGLO5*, has six introns. Comparing the intron–exon display of the glycolate oxidase gene family in tomato and Arabidopsis further highlights the conservation of this gene family among plants—the *GLOs* from the same group showed strong resemblances regarding their corresponding gene structures (Fig. 2).

Motif analysis was conducted with the help of the MEME Suite Program. As a result, a total of 14 distinct motifs were identified. The schematic representation of these motifs, FMN_dh domain, and the phylogenetic tree are provided. The motifs are represented in their relative locations within the proteins. Among the 14 identified motifs, Motif 1 was considered the FMN_dh domain and was uniformly observed across all the *GLO* proteins. All the glycolate oxidase proteins from tomato, Arabidopsis, spinach, and carrot contained Motif 1 and 3. Motif 2 was prevalent exclusively in group A and group B members. Motifs 4, 5, 8, and 11 were observed in almost all the *GLO* proteins except for a few members, while Motif 7 could be observed in all group A and B members. Motif 14 was explicitly present in two spinach glycolate oxidase gene family members (*SoGLO1* and *SoGLO4*), whereas all the twenty *GLOs* harbored Motif 10 except for a group A member. In general, it can be inferred that most of the closely related members in the phylogenetic tree possessed typical motif compositions, suggesting a structural and functional similarity within the genes of the same group (Fig. 3).

3.4 Chromosomal distribution, gene duplication, and synteny analysis

The positions of the *SIGLO* genes were analyzed in the tomato genome and displayed as a map. The results showed that the six *SIGLOs* were randomly distributed on four chromosomes. Three of them, *SIGLO1*, *SIGLO2*, and *SIGLO3*, were located on chromosome 03. The remaining *SIGLOs*, *SIGLO4*, *SIGLO5*, and *SIGLO6*, were reported to be present on chromosome 07, chromosome 08, and chromosome 10, respectively. Thus, chromosome 03 exhibited the maximum number of *SIGLO* genes (Fig. 4a).

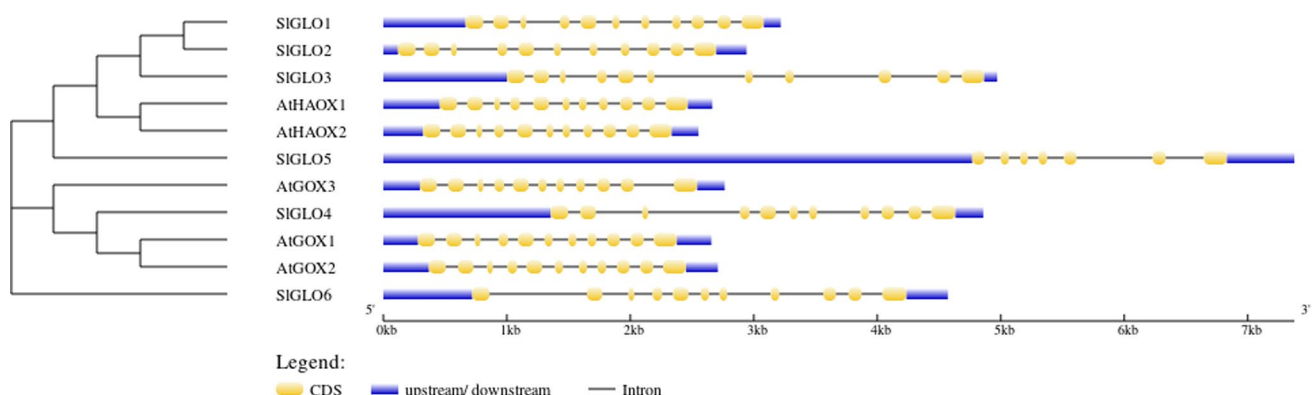


Fig. 2 Gene structure analysis of Arabidopsis and tomato *GLO* genes was carried out using GSDS 2.0. The yellow round rectangles indicate exons (CDS), the blue rectangles indicate upstream and downstream, and the black lines indicate introns



Fig. 3 Distribution of 14 motifs along with the conserved FNN_dh domain in the *GLO* proteins of tomato, Arabidopsis, spinach, and carrot; the motif analysis was carried out using MEME 5.4.1 [1]. Interlinking it with a phylogenetic tree to clearly understand their association, the bars represent motifs with different color codes for various motifs

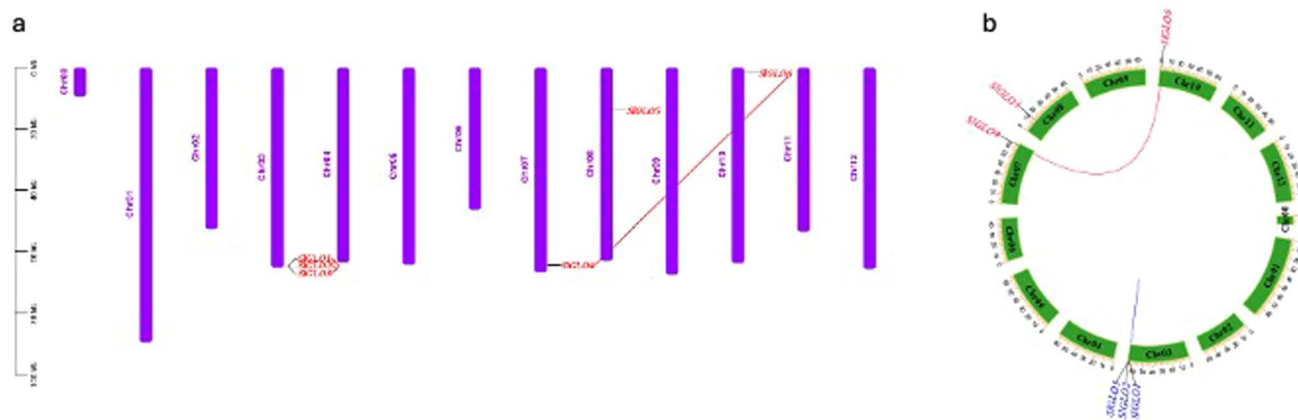


Fig. 4 **a** Distribution of the six *GLO* (*SIGLO*) genes on tomato chromosomes. The approximate location of each *SIGLO* gene is marked with a short black line. Red lines indicate the pairing of genes. **b** Genome-wide synteny analysis of tomato *GLO* genes. Chromosomes are shown in green, and chromosome numbers are displayed in black. The red line indicates the genes resulting from segmental duplication, and the blue line shows those resulting from tandem duplication

Synteny analysis was conducted to analyze the potential duplication events and evolutionary patterns of the *GLO* genes in tomatoes. Four paralogous pairs of *SIGLO* genes were detected. These genes were located either on the same chromosome or on different chromosomes, suggesting that tandem duplication and segmental duplication might have contributed to the expansion of the *GLO* gene family in tomato (Fig. 4a, b).

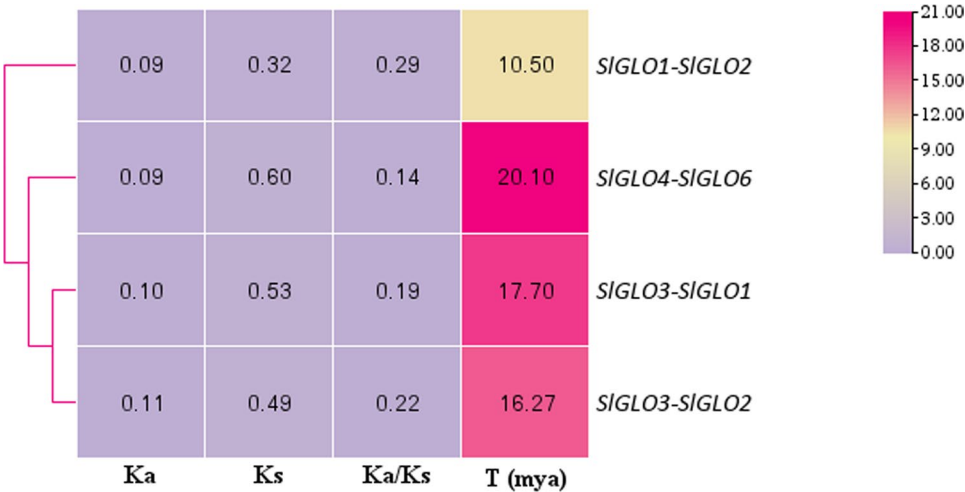
Furthermore, the duplication dates were traced by estimating the Ka and Ks values. The results showed that the Ka/Ks ratio ranged from 0.14 in the *SIGLO4/SIGLO6* pair to 0.29 in the *SIGLO1/SIGLO2* gene pair. The predicted date for segmental gene duplication of the paralogous pair *SIGLO4/SIGLO6* was calculated to be 20.1 mya (million years ago), while for the remaining three paralogous pairs, the tandem duplication date was estimated in the range of 10.5 mya for the *SIGLO1/SIGLO2* gene pair to 17.7 mya for the paralogous pair *SIGLO3*. All four paralogous gene pairs in tomato had a Ka/Ks ratio greater than 0.1 but less than 1, suggesting purifying selective pressure during *GLO* gene family evolution and conservation of functions *SIGLO1* (Fig. 5; Table S2).

The dual synteny analysis results showed that tomato *GLO* genes showed three syntenic gene pairs with Arabidopsis, five with carrots, and one with spinach. Moreover, the details of these gene pairs showed that among the six tomato *GLO* genes, *SIGLO1*, *SIGLO4*, and *SIGLO6* were found in the three comparative syntenic maps, which indicates that these three genes might have played fundamental roles in the *GLO* gene family (Fig. 6).

3.5 Cis-regulatory elements analysis

Several *cis*-acting regulatory elements were found in the promoter regions of each of the six tomato *GLO* genes. The results identified the maximum and minimum CREs in *SIGLO5* and *SIGLO2*, respectively. Other than core *cis*-elements such as CAAT-box and TATA-box, we found a variety of putative CREs that could be divided into various categories based on their known functions, such as hormone-responsive *cis*-regulatory elements, stress-responsive *cis*-elements, and *cis*-elements associated to cellular development. Among the hormonal regulation, cellular growth, and stress response-associated CREs, the stress-responsive *cis*-elements were the most common, and light-responsive *cis*-regulatory elements constituted a large percentage. In addition, some unnamed elements that were randomly distributed in the *SIGLO* genes were also detected. In a total of 75 *cis*-elements obtained as a result, *cis*-elements responsive to different kinds of stress were observed, such as *cis*-elements essential for anaerobic induction (ARE), cold and dehydration-responsive element (DRE core), *cis*-elements involved in drought-inducibility (MBS) and low-temperature responsiveness (LTR), and wound-responsive element (WUN-motif). Other than these, some of the different abiotic and biotic stress-responsive elements (MYB, MYB-like sequence, MYC, Myb, Myc, STRE, TC-rich repeats, as-1, WRE3, and W box) were very common in most of the *SIGLOs*. It has been reported that light-responsive elements (LREs) significantly occur in the genes crucial for light-associated transcriptional activity [9]. In the *SIGLO* promoters, these kinds of elements occur at various frequencies. It included CREs such as those involved in light-responsiveness (3-AF1 binding site, ABRE4, ACE, Box 4, G-Box, G-box and GT1-motif). This *cis*-element was reported to be part of a ‘module’ for light response (AE-box), and CREs that were reported to be parts of other light-responsive elements (chs-CMA2a, GA-motif, GATA-motif, I-box, TCCC-motif, and TCT-motif). Each *SIGLO* gene harbored at least three of these LREs. The category of hormone-responsive elements included *cis*-elements such as abscisic acid-responsive elements (ABRE and ABRE3a), auxin-responsive elements (AuxRR-core and TGA-element), gibberellin-responsive elements (TATC-box, P-box, and

Fig. 5 Time of gene duplication estimated for different paralogous pairs of tomato *GLO* genes based on the Ka and Ks values. Analysis was conducted using the Simple Ka/Ks Calculator (NG) in TBtools, where the Ka/Ks value represents the ratio of nonsynonymous (Ka) to synonymous (Ks) mutations



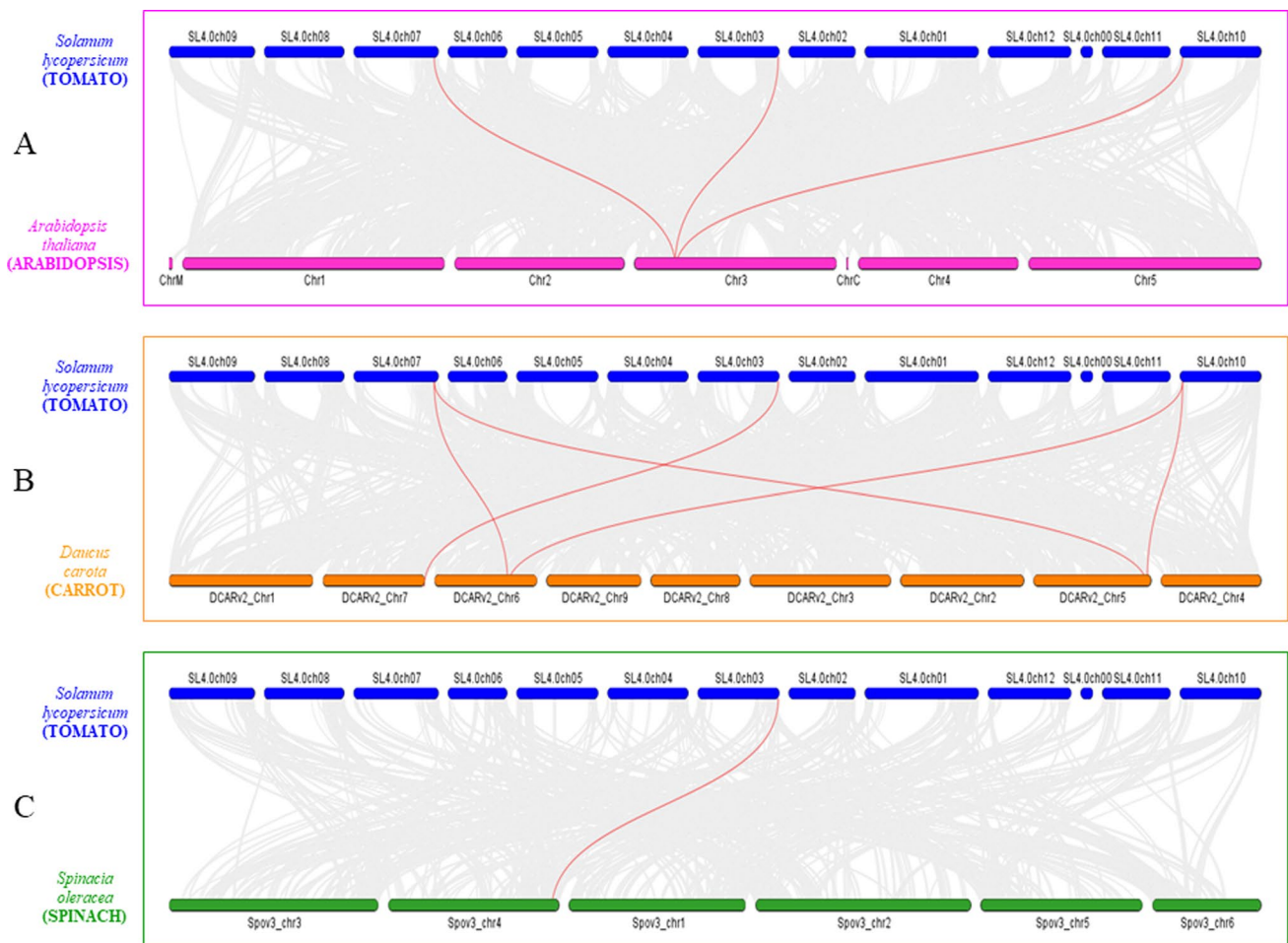


Fig. 6 Analysis of syntenic *GLO* gene pairs between *Solanum lycopersicum* (tomato) and three other plant species, including **A.** *Arabidopsis thaliana*, **B.** *Daucus carota*, and **C.** *Spinacia oleracea*. Gray lines indicate all the collinear blocks in the genome, and red lines indicate the syntenic *GLO* gene pairs. Tomato chromosomes are represented in blue, whereas Arabidopsis, carrot, and spinach are represented in pink, orange, and green, respectively

GARE-motif), ethylene-responsive element (ERE), and *cis*-elements involved in salicylic acid responsiveness (TCA and TCA-element) and methyl jasmonate-responsiveness (CGTCA-motif and TGACG-motif). Each of the six *SIGLOs* contained at least four of these hormone-responsive elements. On the whole, *cis*-elements associated with cellular development were present in the lowest numbers in the tomato glycolate oxidase gene family members; various kinds of CREs were detected, such as those involved in seed-specific regulation (RY-element), meristem expression (CAT-box and CCGTCC-box), endosperm expression (GCN4_motif), zein metabolism regulation (O2-site) and differentiation of the palisade mesophyll cells (HD-Zip 1). Moreover, *cis*-element involved in circadian control (circadian) was found in *SIGLO1*, *SIGLO2*, and *SIGLO3* promoters. In addition to the entire CREs mentioned above, *cis*-elements were also observed, such as (A-box) in *SIGLO3*; (CARE) in *SIGLO5*; (NON) in *SIGLO3*; (OCT) in *SIGLO6* and (AAGAA-motif) in *SIGLO3*, *SIGLO5* and *SIGLO6*. Hence, it can be inferred that the presence of a significant number of *cis*-acting regulatory elements suggests a robust participation of the *SIGLO* gene family in biotic and abiotic stress responses, light-controlled activities, hormone-signaling and growth and development processes (vegetative and reproductive). The distribution of *cis*-regulatory elements identified in the six *GLO* genes of tomato and their functional annotations have been presented (Fig. 7; Table S3).

3.6 Subcellular localization and gene ontology enrichment analysis

The predicted subcellular localization of the *SIGLOs* indicated that three (*SIGLO1*, *SIGLO2*, and *SIGLO3*) were peroxisome. The remaining three (*SIGLO4*, *SIGLO5*, and *SIGLO6*) were situated in cytosol. Members of the glycolate oxidase gene family of the model plant *Arabidopsis* have also been found to be localized to the peroxisomes [28] (Fig. 8).

Fig. 7 Variety of *cis*-regulatory elements identified in the putative *SIGLO* promoters. Color legends indicate the number of *cis* elements in each of the six *SIGLO* genes

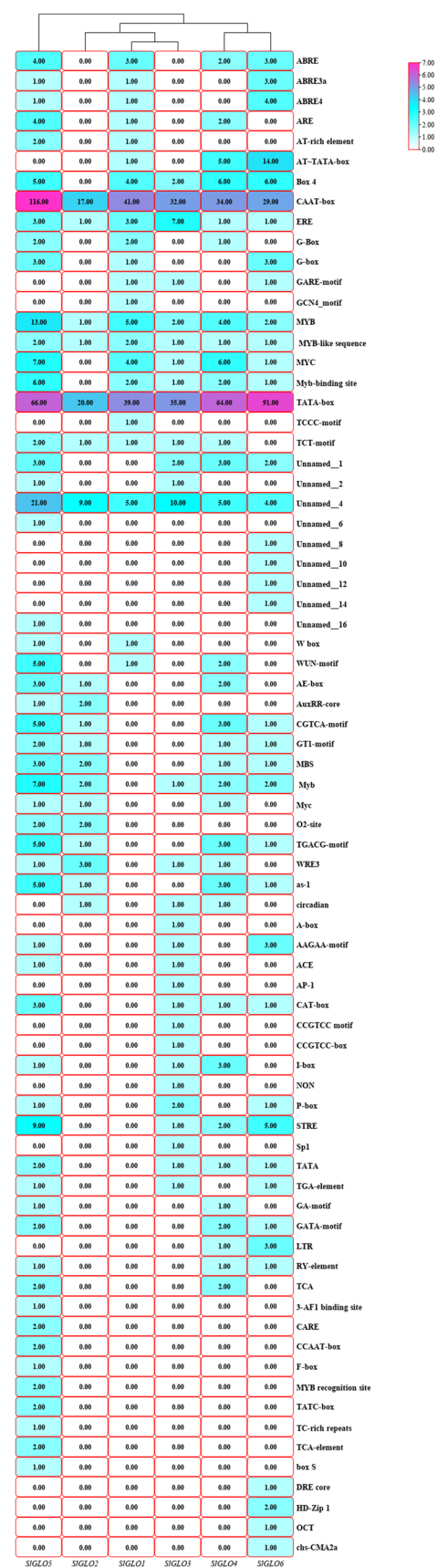
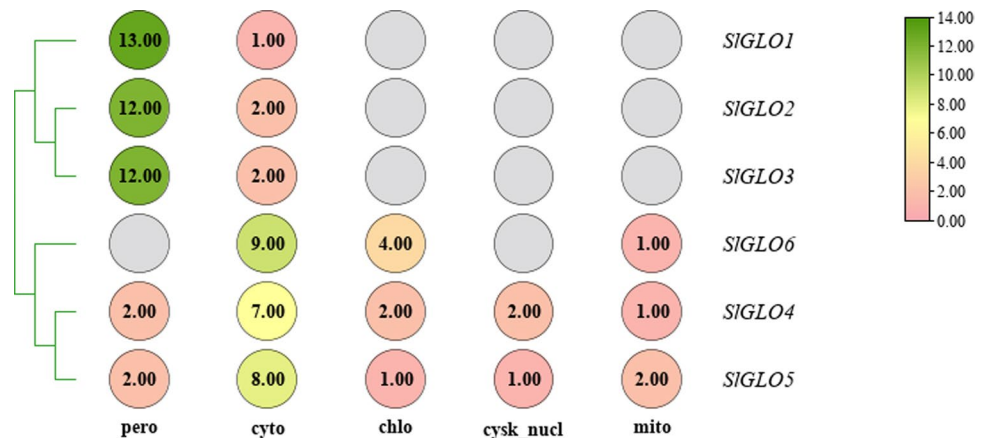


Fig. 8 Subcellular localizations of *SIGLO* genes predicted by employing WoLF PSORT [37]



Gene ontology analysis of the tomato glycolate oxidase gene family members conducted via Phytozome v13 revealed that most of them were involved in molecular functions, including oxidoreductase activity (GO:0016491), and were found to be FMN-binding (GO:0010181). In contrast, biological processes related to these genes were reported to be the oxidation–reduction process (GO:0055114). It could be observed that the gene ontology results were highly consistent with the phylogenetic analysis; *SIGLO* members from group B (*SIGLO1*, *SIGLO2*, and *SIGLO3*) showed similar GO molecular functions such as oxidoreductase activity and FMN-binding. Group A's members, i.e., *SIGLO4* and *SIGLO6*, shared another molecular function, such as catalytic activity, whereas *SIGLO5* from group C exhibited another additional molecular function related to nitronate monooxygenase activity. Moreover, orthologs of the *SIGLO* genes in Arabidopsis were identified against the Arabidopsis genome (Arabidopsis thaliana TAIR10) in the Phytozome database (Table 2).

GO term analysis identified thirteen enriched pathways, each associated with at least one gene from the six *SIGLOs*. Some of them include the hydrogen peroxide biosynthetic process, reactive oxygen species biosynthetic process, photorespiration, defense response to another organism, immune response, response to external biotic stimulus, response to the bacterium, and biological process involved in interspecies interaction between organisms (Fig. 9).

3.7 Gene expression analysis in vegetative and reproductive tissues

The gene expression was analyzed in both vegetative and reproductive tissues. According to the transcriptomic data, all six *SIGLO* genes were expressed in most of the tissues; however, the extent of expression of these genes in each tissue/organ was different from that of the other.

For vegetative tissues, previously generated RNA seq-data was obtained from an experiment designed to study the effects of silencing the gene 'MSH1' (MutS HOMOLOG1) in *Solanum lycopersicum* (cultivar Rutgers). For this purpose, tomato wild-type plants and MSH-1 mutant plants were compared for the phenotypic traits [60, 92]. To investigate the expression profiles of *SIGLO* genes, the gene expression data obtained from the apical meristem tissues of 4-week wild type, dwarf-DR, mild-DR, and epiF3 plants was analyzed and studied (where DR represents 'developmental reprogramming'). The expression patterns have been presented in the form of a Heat map. In a low-to-high range, the lowest expression level was detected for *SIGLO1* (almost undetectable), followed by *SIGLO3* and then *SIGLO2*, which exhibited higher expression levels in the analyzed samples. *SIGLO5* was expressed moderately, whereas *SIGLO4* and *SIGLO6* depicted the highest expressions. *SIGLO4* was expressed to the maximum level from all six genes in all the analyzed apical meristem samples. An important point to be noted here is that an upregulation was observed in the expression of the gene '*SIGLO2*' in two mutant plants compared to the wild type (Fig. 10A; Table S4).

The gene expression profiles in reproductive tissues were investigated and observed to understand the connection between the *GLO* family in tomatoes and plant reproductive growth and development. In an experiment, different phenotypic traits, such as those related to fruit and seed masses, were compared between two nearly isogenic lines of tomato that differed for the fruit mass locus, *fw3.2*. One of the most isogenic lines (NIL) had the large-fruited allele, *fw3.2(ys)*, and the other line had the small-fruited allele, *fw3.2(wt)*. The tomato cultivar 'Yellow Stuffer' was selected, and plants were grown under field and greenhouse conditions. The RNA-seq data of flower buds, pericarp, and seed tissues was collected at various stages [18]. For the gene expression analysis in our study, the RNA-seq data of flower buds, pericarp, and seed was obtained from two experiments available in the Expression Atlas database [60]. Both the

Table 2 Gene Ontology Enrichment Analysis of *S/GLO* genes using Phytozome v1.3 [34]

Gene name	GO annotations		Sub-cellular localization	Gene expression	Ortholog in arabidopsis		Reference
	Molecular functions	Biological process			Gene name	Function	
<i>S/GLO1</i>	FMN binding, oxidoreductase activity	Obsolete oxidation–reduction process	Peroxisome	Seed, red fruits of mycorrhizal plants	<i>AtHAOX1</i>	Plays role in non-photorespiratory metabolism, mainly expressed in seed	[44, 66]
<i>S/GLO2</i>	Oxidoreductase activity, FMN binding	Obsolete oxidation–reduction process	Peroxisome	Apical meristem, flower bud	<i>AtHAOX2</i>	Plays role in non- photorespiratory metabolism, mainly expressed in seed	[44, 66]
<i>S/GLO3</i>	Oxidoreductase activity, fmn binding	Obsolete oxidation–reduction process	Peroxisome	Pericarp, infected root samples under biotic stress	<i>AtGOX2</i>	Involved in photorespiration	[75]
<i>S/GLO4</i>	Oxidoreductase activity, FMN binding, catalytic activity	Obsolete oxidation–reduction process	Cytosol	Apical meristem, flower bud, seed and pericarp, green fruits of non-mycorrhizal plants, infected and un-infected root samples under biotic stress	<i>AtGOX1</i>	Involved in photorespiration	[75]
<i>S/GLO5</i>	FMN binding, oxidoreductase activity, catalytic activity, nitronate monooxygenase activity	Obsolete oxidation–reduction process	Cytosol	Red fruits of mycorrhizal plants apical meristem, seed and pericarp, infected root samples under biotic stress	<i>AtGOX1</i>	Involved in photorespiration	[75]
<i>S/GLO6</i>	Catalytic activity, FMN binding, oxidoreductase activity	Obsolete oxidation–reduction process	Cytosol	Apical meristem, flower bud, pericarp, green fruits of non-mycorrhizal plants, infected root samples under biotic stress	<i>AtGOX3</i>	Supports L-lactate metabolism in roots	[75]

Details about molecular functions and their associated biological processes, sub-cellular localizations, gene expression, and ortholog in Arabidopsis and its putative function

Fig. 9 Gene Ontology analysis of *SIGLO* genes performed at ShinyGO 0.76 [33]. A fold enrichment chart as a dot plot represents the biological processes and several associated *SIGLO* genes

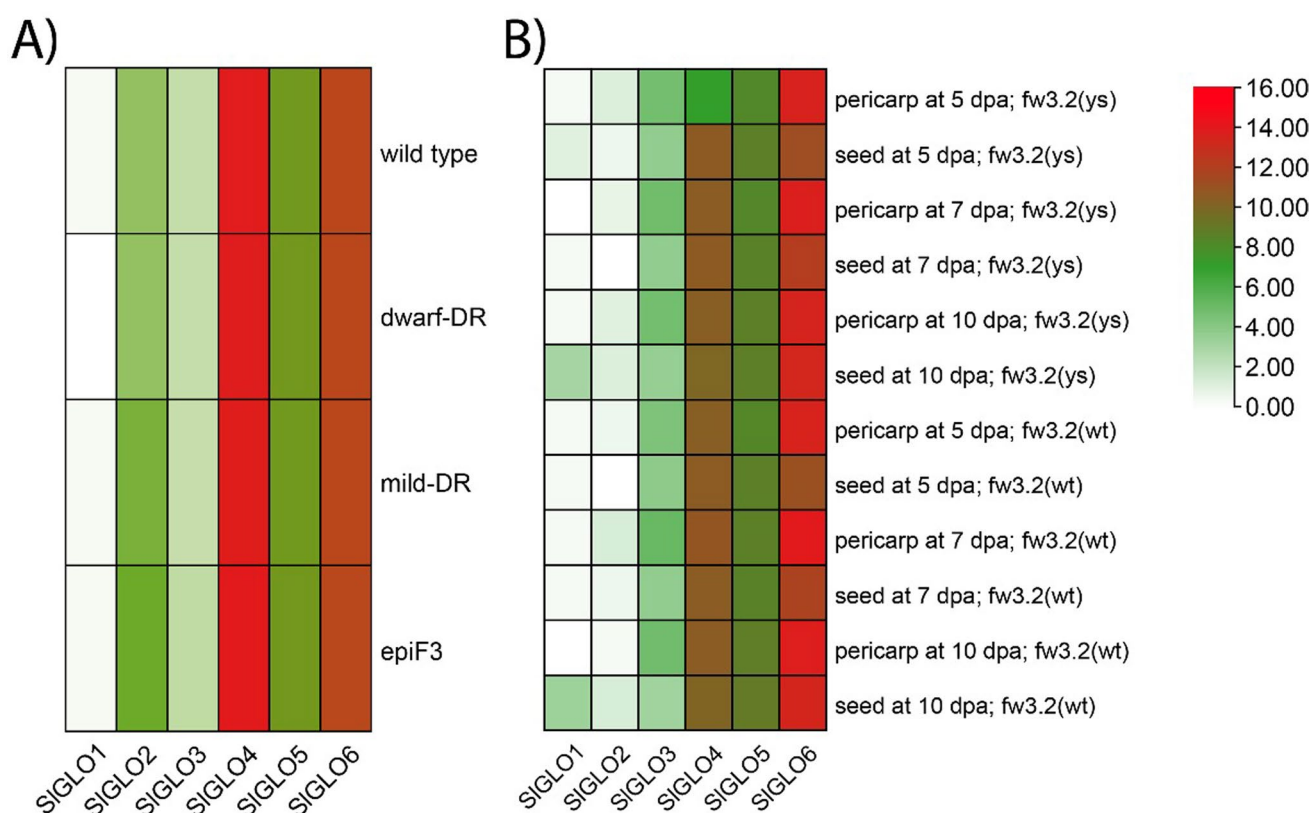
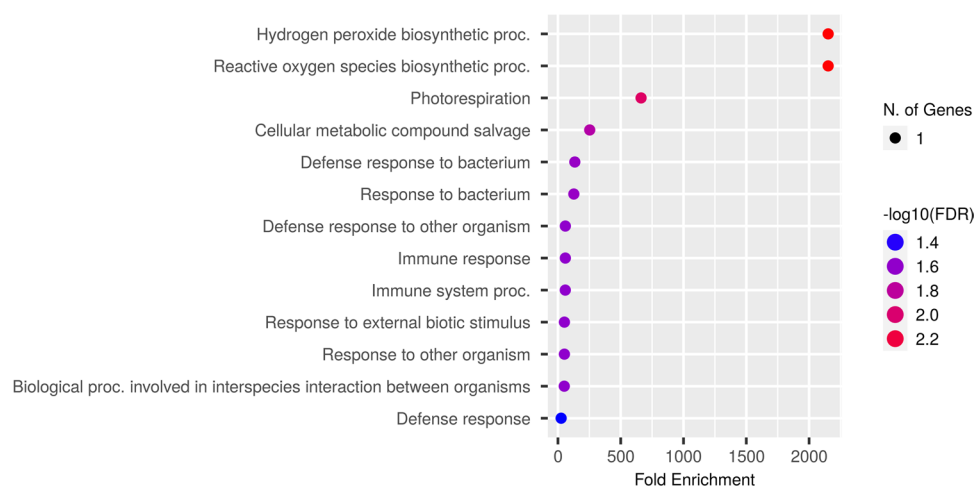


Fig. 10 **A** Heat map showing the expression profile of *SIGLO*s in the apical meristem tissues of four tomato genotypes (wild type and MSH1 mutant plants). The expression of *SIGLO2* was upregulated in two of the mutants, whereas the rest of the genes showed a uniform expression in all the samples. **B** The expression profile of *SIGLO* genes in pericarp and seed, at five dpa (days post-anthesis) and seven dpa (days post-anthesis) in two nearly isogenic lines of tomato differing for the fruit mass locus '*fw3.2*'

experiments were related to the same research/study. The expression profile of flower buds depicted that *SIGLO1* and *SIGLO3* were expressed in a lower range, whereas *SIGLO2* and *SIGLO5* showed moderate expressions. The maximum gene expression level was observed for the two *SIGLO*s, *SIGLO4* and *SIGLO6*. *SIGLO4* represented the highest expression levels in the analyzed flower bud samples (Table S5).

In seed and pericarp tissues, the expression level of *SIGLO1* was slightly higher in seed samples than in the pericarp. While *SIGLO2* and *SIGLO3* exhibited higher expression levels in pericarp tissues than in seed. *SIGLO4* and *SIGLO5* presented varying expression patterns and were expressed moderately in both tissues. *SIGLO6* was the only gene

with significantly higher expressions in the pericarp tissues in all the analyzed samples. Thus, the spatial variations in the expressions of *SIGLO* genes in different tissues indicate that they might be involved in various growth and development processes in tomato plants (Fig. 10B; Table S6).

3.8 Gene expression analysis in tomato fruits of mycorrhizal plants

Previous studies have shown the interaction of hydroponically grown tomato plants with Arbuscular Mycorrhizal Fungi (AMF) improved the nutrient quality of tomato fruits [76]. In the *SIGLO* gene expression, RNA-seq data was derived from an experiment that aimed to study the effects of mycorrhization on the nutritional quality (such as BRIX values and carotenoid contents) of tomato fruits. Picolino cultivar of tomato was cultivated and inoculated with the AMF (*Rhizophagus irregularis*). After a short interval of inoculation, the mycorrhization rates were determined, and RNA-seq was performed for mature green and red fruits of mycorrhizal (treated) and non-mycorrhizal (control) plants [60, 76]. The gene expression data for all six tomato *GLOs* was extracted and analyzed. The results demonstrated that *SIGLO2* expressed at a low level, whereas the expression levels of *SIGLO3* and *SIGLO1* were slightly higher in the red fruits of mycorrhizal plants. *SIGLO6* displayed high expression levels in green fruits and reduced expression in red fruits of both samples (treated and controlled). We observed a maximum expression for *SIGLO4* and *SIGLO5* in all the analyzed fruit samples (Fig. 11; Table S7).

3.9 Gene expression analysis under biotic stress conditions

The gene expression data was obtained from a comparative RNA-seq-based transcriptomic analysis experiment in two tomato cultivars, resistant *Mogeor*, and susceptible *Moneymaker*, to investigate the plant-pathogen interactions. Biotic stress was provided by inoculating young tomato plants with *Pyrenochaeta lycopersici* (for comparison, the control plants were not inoculated with *P. lycopersici*) [56, 60]. The RNA-seq data obtained from the 'root' tissues of infected and control samples of both the cultivars was analyzed to study the expression patterns of *SIGLO* genes in response to biotic stress. We observed that all six *SIGLOs* had shown varied expressions. It was demonstrated that *SIGLO1* and *SIGLO2* showed negligible expression levels in all the samples. *SIGLO3* showed lower-than-moderate expression in the infected root tissues and insignificant expression in the control. The expression level of *SIGLO5* was in a considerable range in all the samples, while *SIGLO4* and *SIGLO6* displayed highly significant expression levels. The maximum expression level was noted for *SIGLO6* in the infected root samples of resistant *Mogeor* cultivar. Since the RNA-seq data in this experiment was obtained at the three-leaf visible stage, the genes exhibiting the minimum expressions (*SIGLO1* and *SIGLO2*) might be expressed at further stages of development (Fig. 12; Table S8).

Fig. 11 Expression profile of *SIGLO* genes in green (early ripening stage) and red (complete ripening stage) fruits of mycorrhizal and non-mycorrhizal tomato plants

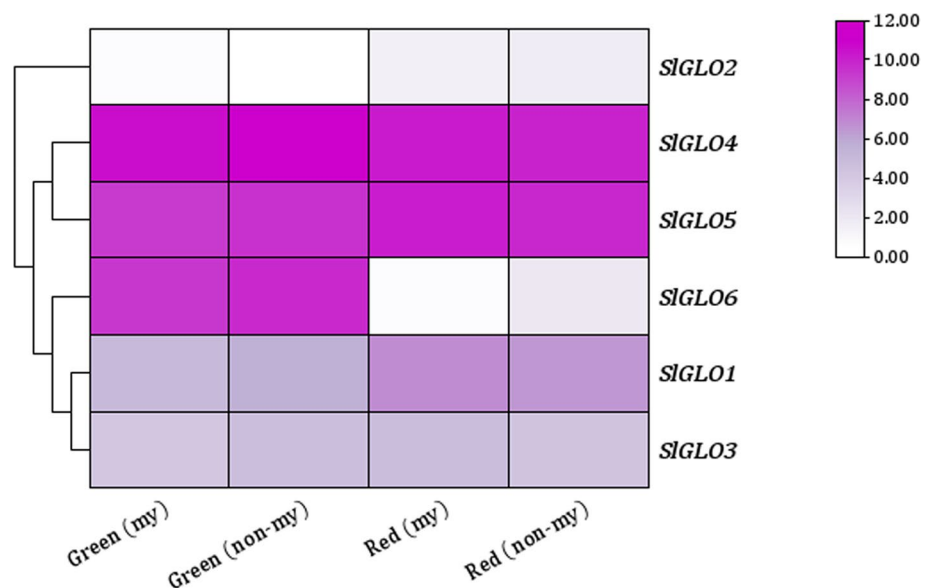
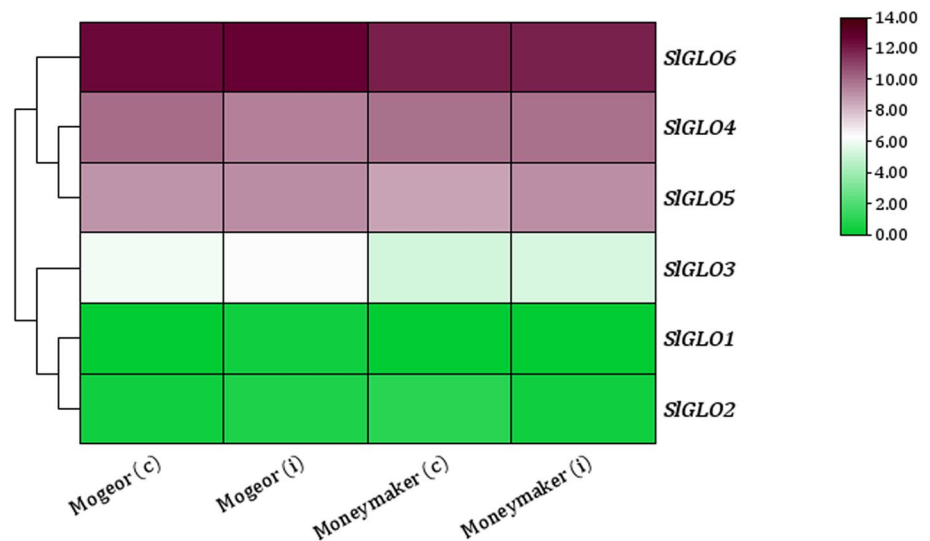


Fig. 12 Heat map showing the expression profile of *SIGLO* genes under biotic stress; expression patterns of *SIGLO*s in infected and control root samples of two genotypes: resistant Mogeor and susceptible Moneymaker



3.10 qRT-PCR expression patterns of *SIGLO* genes in various organs and developmental stages of *C. sativus*

qRT-PCR results showed that *SIGLO* genes were expressed in leaf, stem, anthesis, immature green, orange, and red ripe stages. Specifically, *SIGLO1* was upregulated in orange and was low in anthesis. *SIGLO2* was upregulated in leaf and anthesis but showed low expression in orange and red ripe stages. *SIGLO3* was upregulated in the stem, red ripe, and orange stages while showing low expression in the leaf and immature green stages. *SIGLO4* was upregulated in leaf and anthesis but exhibited low levels in the stem, anthesis, and immature green stages, with little to no expression in the orange and red ripe stages. *SIGLO5* was upregulated in orange, stem, and anthesis, while in immature green and red ripe

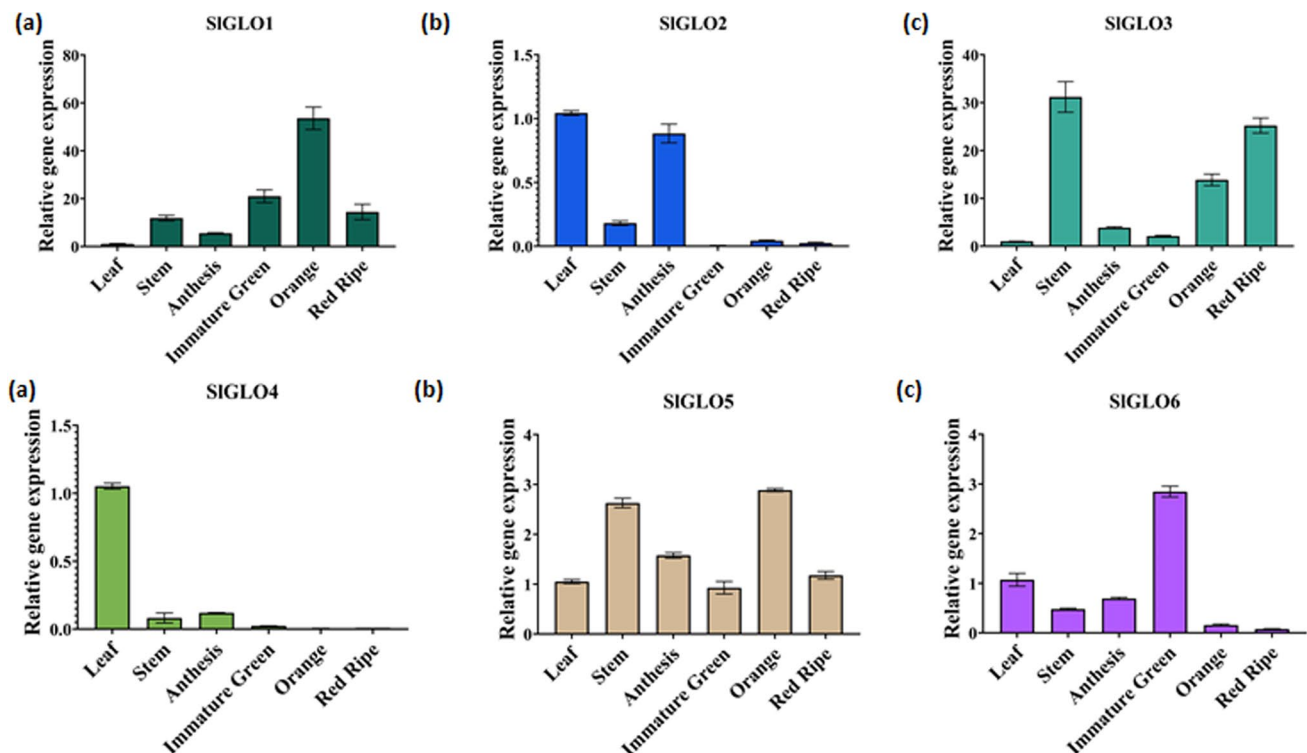


Fig. 13 The qRT-PCR results show the expression levels of six *SIGLO* genes in various organs and developmental stages, including leaf, stem, anthesis, immature green, orange, and red ripe stages

stages, it showed minute expression. *SIGLO6* was upregulated in immature green and leaf stages while exhibiting low levels in orange and red ripe stages (Fig. 13).

3.11 MicroRNA targets prediction

The mature tomato miRNA sequences were retrieved from the Plant miRNA Encyclopedia database (PmiREN2.0) available at (<https://www.pmiren.com/>). To predict the potential targets of tomato miRNAs among the *SIGLO* genes, the identified tomato miRNAs were analyzed using psRNATarget with default parameters, a miRNA target prediction tool for plants. A total of 17 miRNAs targeted five out of the six *SIGLO* genes (Table S9). None of these microRNAs targeted the remaining *SIGLO* gene. The lengths of these miRNAs ranged from 20 to 21 nucleotide bases, and their E-values varied from 4.5 to 5. Among all the *SIGLO*s, *SIGLO4* was the only gene targeted by just one miRNA, while two miRNAs targeted *SIGLO2* *SIGLO5*. Furthermore, *SIGLO1* and *SIGLO3* were targeted by eight and four miRNAs, respectively. *SIGLO6* was reported not to be targeted by any of these miRNAs. While discussing based on groups, *SIGLO*s from group B were targeted the most by these miRNAs, which were targeted by fourteen microRNAs. On the other hand, group A *SIGLO* members were targeted by only one miRNA, which is the least among all the groups, while the single *SIGLO* gene family member from group C was targeted by two miRNAs (Table 3; Table S9).

4 Discussion

This study identified six tomato glycolate oxidase (*SIGLO*) genes based on the conserved FMN domain (Table 1). Interestingly, the *GLO* family is relatively small. The number of *GLO* proteins in *Solanum lycopersicum* is comparable to *Arabidopsis thaliana* and *Oryza sativa*, which have five and four members, respectively [24, 50]. Phylogenetic analysis grouped the *GLO* proteins into three clusters: A, B, and C (Fig. 1). Motif and domain analysis confirmed that all *GLO* family members possess the FMN-dependent dehydrogenase domain, with thirteen additional motifs distributed among the proteins. Group C had the least motifs, while groups A and B featured group-specific motifs like Motif 12 and Motif 13, though not universally present. These conserved motifs may contribute to group-specific functions. Exon–intron analysis revealed that all six *SIGLO* genes contain introns, with exons ranging from 7 to 11 and 6 to 10 (Fig. 2). Groups A and B contained ten introns each, while *SIGLO5* from group C had six. The organization is comparable to *Arabidopsis*' glycolate oxidase gene family, suggesting conservation in structure [47, 96]. Gene duplications, classified as tandem or segmental, were analyzed based on chromosomal locations and synteny (Fig. 4a, b). Among the four paralogous pairs detected, three were tandem duplications, and one was segmental. Ka/Ks ratios for these pairs ranged from 0.14 to 0.29, indicating purifying selection [49, 87]. Duplication events occurred 10.5–20.1 mya (Fig. 5; Table S2), suggesting that tandem duplications primarily expanded the *SIGLO* family, potentially driving functional diversification. Dual synteny analysis revealed the highest number of syntenic gene pairs between tomato and carrot, indicating evolutionary homology (Fig. 6). Cyanobacterial *GLO*-like proteins likely served as the ancestral form, with duplications predating higher plant diversification [29, 45, 79]. Promoter analysis of *SIGLO* genes revealed associations with stress-related mechanisms, hormonal regulation,

Table 3 Information about microRNAs and their target genes; the putative functions of miRNAs have been investigated

miRNA	Target gene	Putative functions of miRNAs	References
miR169	<i>SIGLO1</i>	Involved in regulating tolerance to drought stress	[64]
miRN3070	<i>SIGLO3</i>	Reported to be responsive to mechanical strain exerted on osteoblastic cells (in mice)	[35]
miRN3075	<i>SIGLO2</i> , <i>SIGLO1</i>	Suppresses the migration rate of Schwann cells by targeting contactin 2 (in rat)	[86]
miRN3162	<i>SIGLO3</i>	Plays role in asthma etiopathogenesis by regulating β -Catenin (in mice)	[30]
miR159	<i>SIGLO5</i>	Impacts fruit set and ovule development in tomato	[95]
miR167	<i>SIGLO1</i>	Regulates plant development by controlling the expression levels of <i>ARF6</i> and <i>ARF8</i>	[94]
miRN3103	<i>SIGLO1</i>	Involved in interactions with circRNAs expressed in the mouse LV myocardium (in mice)	[93]
miRN3104	<i>SIGLO4</i>	Reported to regulate the top candidate genes of PHD-TF family	[61]
miRN4423	<i>SIGLO1</i>	Reported to regulate the top candidate genes of PHD-TF family	[61]
miRN3111	<i>SIGLO5</i>	Function unidentified	–
miRN3079	<i>SIGLO2</i>	Function unidentified	–

and development [26, 55]. Hormone-responsive elements included those related to ABA, auxin, and salicylic acid, suggesting a role in hormonal regulatory pathways. Stress-responsive elements, such as MBS, LTR, MYB, STRE, and W-box, were prevalent, indicating involvement in biotic and abiotic stress responses [9, 41]. Light-responsive cis-elements suggest roles in light-mediated activities (Fig. 7; Table S3). GO analysis confirmed FMN binding, crucial for photorespiration, fatty acid oxidation, and stress responses (Table 2) [72]. Expression patterns of *SIGLO* genes were analyzed using the Expression Atlas database. *SIGLO4* and *SIGLO6* showed the highest expression in apical meristem tissues, while *SIGLO1* and *SIGLO3* were minimally expressed. *SIGLO2* displayed a unique expression pattern (Fig. 10; Table S4). These results suggest that *SIGLO* genes may regulate vegetative growth, consistent with reports linking glycolate oxidase to stress responsiveness and metabolic processes [92]. In reproductive tissues, all six *SIGLOs* were expressed. *SIGLO4* and *SIGLO6* exhibited the highest expression in flower bud tissues, with *SIGLO6* also prominent in pericarp tissues. *SIGLO1* showed slight seed expression, while *SIGLO4* was notable in pericarp and seeds. High expression levels of *SIGLO4* and *SIGLO5* in fruit tissues suggest a role in improving nutritional quality (Table S5–S7).

Under stress conditions, *SIGLO* genes showed differential expression in the root tissues of infected plants. Group A genes, particularly *SIGLO6*, were highly expressed in response to *Pyrenochaeta lycopersici* infection, with stronger expression in resistant cultivars (Table S8). This aligns with previous reports linking glycolate oxidase to plant defense mechanisms, where H_2O_2 produced during glycolate oxidation serves as a defense signal [67]. While Arabidopsis and Nicotiana studies highlight glycolate oxidase's positive role in disease resistance, rice studies have noted adverse effects [90]. qRT-PCR results supported these findings, revealing differential expression of *SIGLO* genes across organs and fruit development stages [63]. *SIGLO1* was upregulated during fruit maturation, while *SIGLO2* and *SIGLO4* showed high expression in leaves and reproductive stages, pointing to roles in photosynthesis and early development. *SIGLO3*'s expression in stems and mature fruits suggests a role in structural support and ripening. *SIGLO5* and *SIGLO6*'s expression during early fruit development and photosynthesis highlights their developmental roles [78]. MicroRNA analysis identified 17 miRNAs targeting five of the six *SIGLO* genes (Table 3; Table S9). miR169, known for its role in stress tolerance, targeted *SIGLO1*, suggesting its suppression during stress [64]. miR167 family members targeting ARF6 and ARF8 in Arabidopsis regulate flower maturation, which might similarly apply to tomato *SIGLO* genes [88]. miR3104's targeting of *SIGLO4* may influence histone modification regulation [61]. These findings underscore the potential regulatory roles of miRNAs in *SIGLO* gene expression during growth and stress. This study's findings provide comprehensive insights into the structure, evolution, and functional potential of the *SIGLO* gene family. Functional characterization is essential further to elucidate their roles in development and stress mechanisms.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate The plant material used in this study was obtained by the *Department of Horticulture Faculty of Agricultural Sciences, University of the Punjab, Lahore*. The university approves all material. The rest of the data are publicly available data sets that do not apply to this study.

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

Data transparency Rest assured, I have ensured that all data, materials, software applications, and custom code supporting the claims made in this article are in full compliance with field standards. I have considered the possibility of individual journal policies regarding research data sharing, considering the norms and expectations of our discipline.

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References

- Ahammed GJ, Li X, Zhang G, Zhang H, Shi J, Pan C, Yu J, Shi K. Tomato photorespiratory glycolate-oxidase-derived H₂O₂ production contributes to basal defence against *Pseudomonas syringae*. *Plant Cell Environ*. 2018;41:1126–38.
- Ahiakpa JK, Magdy M, Karikari B, Munir S, Mumtaz MA, Tamim SA, Mahmood S, Liu G, Chen W, Wang Y. Genome-wide identification and expression profiling of tomato invertase genes indicate their response to stress and phytohormones. *J Plant Growth Regul*. 2022;41:1481–98.
- Ahmad A, Han S, Sami A, Haider MZ, Ali Q, Shafiq M, Ali D, Iqbal J, Manzoor MA, Sabir IA, Wang Y. Genome-wide identification of Cytochrome P450 gene in Fall Armyworm (*Spodoptera frugiperda*) in response to insecticide resistance. *Plant Stress*. 2024;14:100579.
- Ahmad A, Sami A, Habib U, Ali M, Shafiq M, Haider MZ, Ahmad S, Ali B, Harakeh S, Makki RM, Chaudhary T, Soudy FA. Genome wide analysis of carotenoid cleavage oxygenases (CCO) gene family in *Arachis hypogaea* (peanut) under biotic stress. *BMC Genom*. 2024;25:469.
- Akram J, Haider MZ, Shafiq M, Sami A, Manzoor MT, Ali S, Haider MS, Shahid MA, Siddique R. Cell signaling response under plants stress. In: Shahid M, Gaur R, editors. *Molecular dynamics of plant stress and its management*. Singapore: Springer; 2024. p. 447–74.
- Ali M, Shafiq M, Haider MZ, Sami A, Alam P, Albalawi T, Kamran Z, Sadiq S, Hussain M, Shahid MA, Jeridi M, Ashraf GA, Manzoor MA, Sabir IA. Genome-wide analysis of NPR1-like genes in citrus species and expression analysis in response to citrus canker (*Xanthomonas axonopodis* pv. citri). *Front Plant Sci*. 2024;15:1333286.
- Ali Q, Sami A, Haider MZ, Ashfaq M, Javed MA. Antioxidant production promotes defense mechanism and different gene expression level in *Zea mays* under abiotic stress. *Sci Rep*. 2024;14:7114.
- Amjad M, Wang Y, Han S, Haider MZ, Sami A, Batool A, Shafiq M, Ali Q, Dong J, Sabir IA, Manzoor MA. Genome wide identification of phenylalanine ammonia-lyase (PAL) gene family in *Cucumis sativus* (cucumber) against abiotic stress. *BMC Genom Data*. 2024;25:76.
- Arefin S, Bhuiyan MFH, Akther J, Prodhan SH, Hoque H. The dynamics of cis-regulatory elements in promoter regions of tomato sucrose transporter genes. *Int J Veget Sci*. 2021;27:167–86.
- Ayuso-Fernández I, Emrich-Mills TZ, Haak J, Golten O, Hall KR, Schwaiger L, Moe TS, Stepanov AA, Ludwig R, Cutsail Iii GE. Mutational dissection of a hole hopping route in a lytic polysaccharide monooxygenase (LPMO). *Nat Commun*. 2024;15:3975.
- Bailey TL, Johnson J, Grant CE, Noble WS. The MEME suite. *Nucl Acids Res*. 2015;43:W39–49.
- Bernhofer M, Goldberg T, Wolf S, Ahmed M, Zaugg J, Boden M, Rost B. NLSdb—major update for database of nuclear localization signals and nuclear export signals. *Nucl Acids Res*. 2018;46:D503–8.
- Bhatti MHT, Han S, Sami A, Haider MZ, Shafiq M, Ali M, Ahmad S, Ali Q, Sabir IA, Manzoor MA. Genome-wide analysis of XTH Gene Family in Cucumber (*Cucumis sativus*) against different insecticides to enhance defense mechanism. *Plant Stress*. 2024;13:100538.
- Bhatti MHT, Sami A, Haider MZ, Shafiq M, Naeem S, Tariq MR, Ahmad S, Irfan U. Genetic diversity of vegetable crops and utilization in food and nutritional security. In: Al-Khayri JM, Jain SM, Penna S, editors. *Sustainable utilization and conservation of plant genetic diversity*. Singapore: Springer; 2024. p. 171–97.
- Bohman S, Wang M, Dixelius C. Arabidopsis thaliana-derived resistance against *Leptosphaeria maculans* in a *Brassica napus* genomic background. *Theor Appl Genet*. 2002;105:498–504.
- Cai X, Ge C, Xu C, Wang X, Wang S, Wang Q. Expression analysis of oxalate metabolic pathway genes reveals oxalate regulation patterns in spinach. *Molecules*. 2018;23:1286.
- Cai X, Zhang Y, Zhang C, Zhang T, Hu T, Ye J, Zhang J, Wang T, Li H, Ye Z. Genome-wide analysis of plant-specific Dof transcription factor family in tomato. *J Integr Plant Biol*. 2013;55:552–66.
- Chakrabarti M, Zhang N, Sauvage C, Muñoz S, Blanca J, Cañizares J, Diez MJ, Schneider R, Mazourek M, McClead J. A cytochrome P450 regulates a domestication trait in cultivated tomato. *Proc Natl Acad Sci*. 2013;110:17125–30.
- Chao WS, Li X, Horvath DP, Anderson JV. Genetic loci associated with freezing tolerance in a European rapeseed (*Brassica napus* L.) diversity panel identified by genome-wide association mapping. *Plant Direct*. 2022;6:e405.
- Chen C, Chen H, Zhang Y, Thomas HR, Frank MH, He Y, Xia R. TBtools: an integrative toolkit developed for interactive analyses of big biological data. *Mol Plant*. 2020;13:1194–202.
- Cheng F, Wu J, Fang L, Wang X. Syntenic gene analysis between *Brassica rapa* and other Brassicaceae species. *Front Plant Sci*. 2012;3:198.
- Cui L-L, Lu Y-S, Li Y, Yang C, Peng X-X. Overexpression of glycolate oxidase confers improved photosynthesis under high light and high temperature in rice. *Front Plant Sci*. 2016;7:1165.
- Dai X, Zhuang Z, Zhao PX. psRNATarget: a plant small RNA target analysis server (2017 release). *Nucl Acids Res*. 2018;46:W49–54.
- Dellero Y, Jossier M, Glab N, Oury C, Tcherkez G, Hodges M. Decreased glycolate oxidase activity leads to altered carbon allocation and leaf senescence after a transfer from high CO₂ to ambient air in *Arabidopsis thaliana*. *J Exp Bot*. 2016;67:3149–63.
- Dellero Y, Mauve C, Boex-Fontvieille E, Flesch V, Jossier M, Tcherkez G, Hodges M. Experimental evidence for a hydride transfer mechanism in plant glycolate oxidase catalysis. *J Biol Chem*. 2015;290:1689–98.
- Ding Y, Zhu J, Zhao D, Liu Q, Yang Q, Zhang T. Targeting cis-regulatory elements for rice grain quality improvement. *Front Plant Sci*. 2021;12:705834.
- Edgar RC. MUSCLE: multiple sequence alignment with high accuracy and high throughput. *Nucl Acids Res*. 2004;32:1792–7.
- Engqvist MK, Schmitz J, Gertzmann A, Florian A, Jaspert N, Arif M, Balazadeh S, Mueller-Roeber B, Fernie AR, Maurino VG. GLYCOLATE OXIDASE3, a glycolate oxidase homolog of yeast L-lactate cytochrome c oxidoreductase, supports L-lactate oxidation in roots of *Arabidopsis*. *Plant Physiol*. 2015;169:1042–61.

29. Esser C, Kuhn A, Groth G, Lercher MJ, Maurino VG. Plant and animal glycolate oxidases have a common eukaryotic ancestor and convergently duplicated to evolve long-chain 2-hydroxy acid oxidases. *Mol Biol Evol*. 2014;31:1089–101.
30. Fang C, Lu W, Li C, Peng X, Wang Y, Huang X, Yao Z, Cai N, Huang Y, Zhang X. MiR-3162-3p is a novel microRNA that exacerbates asthma by regulating β -catenin. *PLoS ONE*. 2016;11: e0149257.
31. Gao M, Han X, Yang W, Sun H, Zhang L, Xu H. A strategy for improving saline-alkali soil properties and cotton stress tolerance using vitamin C industrial wastes: a “prebiotics-probiotics” interaction between organic acids and *Bacillus endophyticus*. *Ind Crops Prod*. 2024;220: 119187.
32. Gasteiger E, Hoogland C, Gattiker A, Wilkins MR, Appel RD, Bairoch A. Protein identification and analysis tools on the ExPASy server. *Proteom Protoc Handb*. 2005; 571–607.
33. Ge SX, Jung D, Yao R. ShinyGO: a graphical gene-set enrichment tool for animals and plants. *Bioinformatics*. 2019;36:2628–9.
34. Goodstein DM, Shu S, Howson R, Neupane R, Hayes RD, Fazo J, Mitros T, Dirks W, Hellsten U, Putnam N. Phytozome: a comparative platform for green plant genomics. *Nucl Acids Res*. 2012;40:D1178–86.
35. Guo Y, Wang Y, Liu Y, Liu Y, Zeng Q, Zhao Y, Zhang X, Zhang X. MicroRNA-218, microRNA-191*, microRNA-3070a and microRNA-33 are responsive to mechanical strain exerted on osteoblastic cells. *Mol Med Rep*. 2015;12:3033–8.
36. Haider MZ, Sami A, Shafiq M, Anwar W, Ali S, Ali Q, Muhammad S, Manzoor I, Shahid MA, Ali D. Genome-wide identification and in-silico expression analysis of carotenoid cleavage oxygenases gene family in *Oryza sativa* (rice) in response to abiotic stress. *Front Plant Sci*. 2023;14:1269995.
37. Horton P, Park K-J, Obayashi T, Fujita N, Harada H, Adams-Collier CJ, Nakai K. WoLF PSORT: protein localization predictor. *Nucl Acids Res*. 2007;35:W585–7.
38. Hu B, Jin J, Guo A-Y, Zhang H, Luo J, Gao G. GSDS 2.0: an upgraded gene feature visualization server. *Bioinformatics*. 2015;31:1296–7.
39. Hussain M, Javed MM, Sami A, Shafiq M, Ali Q, Mazhar HS-U-D, Tabassum J, Javed MA, Haider MZ, Hussain M, Sabir IA, Ali D. Genome-wide analysis of plant specific YABBY transcription factor gene family in carrot (*Dacus carota*) and its comparison with Arabidopsis. *BMC Genom Data*. 2024;25:26.
40. Ishfaq Q, Sami A, Zeshan Haider M, Ahmad A, Shafiq M, Ali Q, Batool A, Haider MS, Ali D, Alam P. Genome wide identification of the NPR1 Gene family in plant defense mechanisms against biotic stress in chili (*Capsicum annuum* L.). *Front Microbiol*. 2024;15:1437553.
41. Islam MA, Rahman MM, Rahman MM, Jin X, Sun L, Zhao K, Wang S, Sikdar A, Noor H, Jeon J-S. In silico and transcription analysis of trehalose-6-phosphate phosphatase gene family of wheat: trehalose synthesis genes contribute to salinity, drought stress and leaf senescence. *Genes*. 2021;12:1652.
42. Jossier M, Oury C, Glab N, Hodges M. Purification of recombinant N-terminal histidine-tagged arabidopsis thaliana phosphoglycolate phosphatase 1, glycolate oxidase 1 and 2, and hydroxypyruvate reductase 1. In: *Photorespiration: methods and protocols*. Springer; 2024. p. 97–111.
43. Kaul T, Ramkumar M, Bharti J, Gupta I, Chakraborty P, Thangaraj A, Verma R, Kaul R, Nehra M. Unlocking the micronutrient riches of *Moringa Oleifera*: a comparative multi-omics study investigating the differential control of genes involved in oxalate metabolism. *Moringa Oleifera*. 2025.
44. Kerchev P, Waszczak C, Lewandowska A, Willems P, Shapiguzov A, Li Z, Alseekh S, Mühlenbock P, Hoeberichts FA, Huang J, Van Der Kelen K, Kangasjärvi J, Fernie AR, De Smet R, Van De Peer Y, Messens J, Van Breusegem F. Lack of GLYCOLATE OXIDASE1, but Not GLYCOLATE OXIDASE2, attenuates the photorespiratory phenotype of CATALASE2-deficient arabidopsis. *Plant Physiol*. 2016;171:1704–19.
45. Kern R, Facchinelli F, Delwiche C, Weber AP, Bauwe H, Hagemann M. Evolution of photorespiratory glycolate oxidase among Archaeplastida. *Plants*. 2020;9:106.
46. Launay A, Jolivet S, Clément G, Zarattini M, Dellero Y, Le Hir R, Jossier M, Hodges M, Expert D, Fagard M. DspA/E-triggered non-host resistance against *E. amylovora* depends on the Arabidopsis GLYCOLATE OXIDASE 2 gene. *Int J Mol Sci*. 2022;23:4224. <https://doi.org/10.3390/ijms23084224>.
47. Lecharny A, Boudet N, Gy I, Aubourg S, Kreis M. Introns in, introns out in plant gene families: a genomic approach of the dynamics of gene structure. *J Struct Funct Genom*. 2003;3:111–6.
48. Lescot M, Déhais P, Thijs G, Marchal K, Moreau Y, Van De Peer Y, Rouzé P, Rombauts S. PlantCARE, a database of plant cis-acting regulatory elements and a portal to tools for in silico analysis of promoter sequences. *Nucl Acids Res*. 2002;30:325–7.
49. Li H, Liu L, Qiu Z, He F, Dong W. Complete mitochondrial genome assembly and comparative analysis of *Colocasia esculenta*. *BMC Plant Biol*. 2024;25:67.
50. Li X, Liao M, Huang J, Xu Z, Lin Z, Ye N, Zhang Z, Peng X. Glycolate oxidase-dependent H₂O₂ production regulates IAA biosynthesis in rice. *BMC Plant Biol*. 2021;21:1–14.
51. Li X, Tang H, Xu T, Wang P, Ma F, Wei H, Fang Z, Wu X, Wang Y, Xue Y. N-terminal acetylation orchestrates glycolate-mediated ROS homeostasis to promote rice thermoresponsive growth. *New Phytol*. 2024;243:1742–57.
52. Lindqvist Y. Refined structure of spinach glycolate oxidase at 2 Å resolution. *J Mol Biol*. 1989;209:151–66.
53. Liu Y, Wu W, Chen Z. Structures of glycolate oxidase from *Nicotiana benthamiana* reveal a conserved pH sensor affecting the binding of FMN. *Biochem Biophys Res Commun*. 2018;503:3050–6.
54. Lu S, Wang J, Chitsaz F, Derbyshire MK, Geer RC, Gonzales NR, Gwadz M, Hurwitz DI, Marchler GH, Song JS. CDD/SPARCLE: the conserved domain database in 2020. *Nucl Acids Res*. 2020;48:D265–8.
55. Ma Z, Hu L, Jiang W. Understanding AP2/ERF transcription factor responses and tolerance to various abiotic stresses in plants: a comprehensive review. *Int J Mol Sci*. 2024;25:893.
56. Milc J, Bagnaresi P, Aragona M, Valente M, Biselli C, Infantino A, Francia E, Pecchioni N. Comparative transcriptome profiling of the response to *Pyrenochaeta lycopersici* in resistant tomato cultivar Mogeor and its background genotype—susceptible MoneyMaker. *Funct Integr Genom*. 2019;19:811–26.
57. Mistry J, Chuguransky S, Williams L, Qureshi M, Salazar GA, Sonnhammer EL, Tosatto SC, Paladin L, Raj S, Richardson LJ. Pfam: the protein families database in 2021. *Nucl Acids Res*. 2021;49:D412–9.

58. Naeem S, Wang Y, Han S, Haider MZ, Sami A, Shafiq M, Ali Q, Bhatti MHT, Ahmad A, Sabir IA, Dong J, Alam P, Manzoor MA. Genome-wide analysis and identification of Carotenoid Cleavage Oxygenase (CCO) gene family in coffee (*Coffea arabica*) under abiotic stress. BMC Genomic Data. 2024;25:71.
59. Najafi M, Nasr-Esfahani M, Vatandoost J, Hassanzade-Khankahdani H, Moeini MJ. Transcriptome-based analysis of candidate gene markers associated with resistance mechanism to *Phytophthora melonis* that causes root and crown rot in pumpkin. Funct Plant Biol. 2024;51:FP23038.
60. Papatheodorou I, Fonseca NA, Keays M, Tang YA, Barrera E, Bazant W, Burke M, Füllgrabe A, Fuentes AM-P, George N, Huerta L, Koskinen S, Mohammed S, Geniza M, Preece J, Jaiswal P, Jarnuczak AF, Huber W, Stegle O, Vizcaino JA, Brazma A, Petryszak R. Expression Atlas: gene and protein expression across multiple studies and organisms. Nucl Acids Res. 2017;46:D246–51.
61. Pathania S, Kumar M. Computational identification of key genes and regulatory miRNAs responsible for differential regulation of host-pathogen response during early and late blight diseases in tomato. Plant Stress. 2022;5:100098.
62. Popov VN, Dmitrieva EA, Eprintsev AT, Igamberdiev AU. Glycolate oxidase isoforms are distributed between the bundle sheath and mesophyll tissues of maize leaves. J Plant Physiol. 2003;160:851–7.
63. Poyatos-Pertíñez S, Quinet M, Ortiz-Atienza A, Yuste-Lisbona FJ, Pons C, Giménez E, et al. A Factor Linking Floral Organ Identity and Growth Revealed by Characterization of the Tomato Mutant unfinished flower development (ufd). Front Plant Sci. 2016;7:1648. <https://doi.org/10.3389/fpls.2016.01648>.
64. Rao S, Balyan S, Jha S, Mathur S. Novel insights into expansion and functional diversification of MIR169 family in tomato. Planta. 2020;251:1–17.
65. Rasheed MU, Malik A, Haider MZ, Sami A, Shafiq M, Ali Q, Javed MA, Ali A. NPR1-like genes in *Theobroma cacao*: evolutionary insights and potential in enhancing resistance to *Phytophthora megakarya*. PLoS ONE. 2025;20: e0318506.
66. Reumann S, Ma C, Lemke S, Babujee L, AraPerox. A database of putative Arabidopsis proteins from plant peroxisomes. Plant Physiol. 2004;136:2587–608.
67. Rojas CM, Senthil-Kumar M, Wang K, Ryu C-M, Kaundal A, Mysore KS. Glycolate oxidase modulates reactive oxygen species-mediated signal transduction during nonhost resistance in *Nicotiana benthamiana* and Arabidopsis. Plant Cell. 2012;24:336–52.
68. Sami A, Haider MZ, Bhatti MHT, Batool A, Shahid MA, Shafiq M. Delivery systems for plant genome editing technology. In: Al-Khayri JM, Ingle KP, Jain SM, Penna S, editors. Plant molecular breeding in genomics era: applications. Cham: Springer; 2024. p. 315–49.
69. Sami A, Haider MZ, Shafiq M. Microbial nanoenzymes: features and applications. In: Fungal secondary metabolites. Elsevier; 2024b, p. 353–67.
70. Sami A, Haider MZ, Shafiq M, Sadiq S, Ahmad F. Genome-wide identification and in-silico expression analysis of CCO gene family in sunflower (*Helianthus annuus*) against abiotic stress. Plant Mol Biol. 2024;114:1–19.
71. Sami A, Han S, Haider MZ, Khizar R, Ali Q, Shafiq M, Tabassum J, Khalid MN, Javed MA, Sajid M, Manzoor MA, Sabir IA. Genetics aspect of vitamin C (Ascorbic acid) biosynthesis and signaling pathways in fruits and vegetables crops. Funct Integr Genom. 2024;24:73.
72. Sandoval FJ, Zhang Y, Roje S. Flavin nucleotide metabolism in plants: monofunctional enzymes synthesize FAD in plastids. J Biol Chem. 2008;283:30890–900.
73. Sarwar S, Sami A, Haider MZ, Tasawar L, Akram J, Ahmad A, Shafiq M, Zaki HEM, Ondrasek G, Shahid MS. Genome-wide identification and in silico expression analysis of CCO gene family in *Citrus clementina* (Citrus) in response to abiotic stress. Plants. 2025;14:249.
74. Schäfer P, Hückelhoven R, Kogel K-H. The white barley mutant albostrans shows a supersusceptible but symptomless interaction phenotype with the hemibiotrophic fungus *Bipolaris sorokiniana*. Mol Plant Microbe Interact. 2004;17:366–73.
75. Schmitz J, Hüdig M, Meier D, Linka N, Maurino VG. The genome of *Ricinus communis* encodes a single glycolate oxidase with different functions in photosynthetic and heterotrophic organs. Planta. 2020;252:1–12.
76. Schubert R, Werner S, Cirka H, Rödel P, Tandron Moya Y, Mock H-P, Hutter I, Kunze G, Hause B. Effects of arbuscular mycorrhization on fruit quality in industrialized tomato production. Int J Mol Sci. 2020;21:7029.
77. Shafiq M, Manzoor M, Bilal M, Manzoor T, Anees MM, Rizwan M, Haider MZ, Sami A, Haider MS. Genome-wide analysis of plant specific YABBY transcription factor gene family in watermelon (*Citrullus lanatus*) and Arabidopsis. J Appl Res Plant Sci. 2024;5:63–78.
78. Sharma P, Thakur S, Negi R. Recent advances in breeding of tomato- A review. Int J Curr Microbiol Appl Sci. 2019;8(03):1275–83. <https://doi.org/10.20546/ijcmas.2019.803.151>.
79. Shriti S, Bhar A, Roy A. Unveiling the role of epigenetic mechanisms and redox signalling in alleviating multiple abiotic stress in Plants. Front Plant Sci. 2024;15:1456414.
80. Sugri I, Akansake D, Abukari IA, Low JW, Abidin PE, Dery EK, Acheremu K, Adjebeng-Danquah J, Bidzakin JK, Abubakari M. Production innovations to enhance sweetpotato as a food security crop in northern Ghana. Crop Sci. 2024;64:1452–70.
81. Sun H, Ren M, Zhang J. Genome-wide identification and expression analysis of fibrillin (FBN) gene family in tomato (*Solanum lycopersicum* L.). PeerJ. 2022;10:e13414.
82. Sun Y-G, He Y-Q, Wang H-X, Jiang J-B, Yang H-H, Xu X-Y. Genome-wide identification and expression analysis of GDSL esterase/lipase genes in tomato. J Integr Agric. 2022;21:389–406.
83. Taler D, Galperin M, Benjamin I, Cohen Y, Kenigsbuch D. Plant eR genes that encode photorespiratory enzymes confer resistance against disease. Plant Cell. 2004;16:172–84.
84. Tamura K, Stecher G, Kumar S. MEGA11: molecular evolutionary genetics analysis version 11. Mol Biol Evol. 2021;38:3022–7.
85. Umeohia UE, Olapade AA. Quality attributes, physiology, and postharvest technologies of tomatoes (*Lycopersicon esculentum*)—A Review. Am J Food Sci Tech. 2024;12:42–64.
86. Wang P, He J, Wang S, Wang X, Liu Q, Peng W, Qian T. miR-3075 inhibited the migration of Schwann cells by targeting Cntn2. Neurochem Res. 2018;43:1879–86.
87. Wen B, Zhang F, Wu X, Li H. Characterization of the tomato (*Solanum lycopersicum*) pectin methylesterases: evolution, activity of isoforms and expression during fruit ripening. Front Plant Sci. 2020;11:238.
88. Wu M-F, Tian Q, Reed JW. Arabidopsis microRNA167 controls patterns of ARF6 and ARF8 expression, and regulates both female and male reproduction. Development. 2006;133(21):4211–8.

89. Xu H, Zhang J, Zeng J, Jiang L, Liu E, Peng C, He Z, Peng X. Inducible antisense suppression of glycolate oxidase reveals its strong regulation over photosynthesis in rice. *J Exp Bot.* 2009;60:1799–809.
90. Xu Y-P, Yang J, Cai X-Z. Glycolate oxidase gene family in *Nicotiana benthamiana*: genome-wide identification and functional analyses in disease resistance. *Sci Rep.* 2018;8:1–11.
91. Yang M, Li Z, Zhang K, Zhang X, Zhang Y, Wang X, Han C, Yu J, Xu K, Li D. Barley stripe mosaic virus yb interacts with glycolate oxidase and inhibits peroxisomal ROS production to facilitate virus infection. *Mol Plant.* 2018;11:338–41.
92. Yang X, Kundariya H, Xu Y-Z, Sandhu A, Yu J, Hutton SF, Zhang M, Mackenzie SA. MutS HOMOLOG1-derived epigenetic breeding potential in tomato. *Plant Physiol.* 2015;168:222–32.
93. Yang Y, Chen H, Ding N, Wang S, Duan Z, Birnbaum Y, Ye Y, Qian J. Expression profiling of circular RNAs and microRNAs in heart tissue of mice with alcoholic cardiomyopathy. *Cell Physiol Biochem.* 2018;46:2284–96.
94. Yao X, Chen J, Zhou J, Yu H, Ge C, Zhang M, Gao X, Dai X, Yang Z-N, Zhao Y. An essential role for miRNA167 in maternal control of embryonic and seed development. *Plant Physiol.* 2019;180:453–64.
95. Zhao P, Wang F, Deng Y, Zhong F, Tian P, Lin D, Deng J, Zhang Y, Huang T. Sly-miR159 regulates fruit morphology by modulating GA biosynthesis in tomato. *Plant Biotechnol J.* 2022;20:833–45.
96. Zhou Y, Hu L, Ye S, Jiang L, Liu S. Genome-wide identification of glutathione peroxidase (GPX) gene family and their response to abiotic stress in cucumber. *3 Biotech.* 2018;8:1–11.

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